

ARANYAPRATHET, Thailand

WMO: 484620

Lat: 13.6886N

Lon: 102.5042E

Elev: 49

StdP: 100.74

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	18.1	19.5	11.5	8.5	22.1	13.0	9.4	22.9	6.2	26.5	5.5	26.4	0.5	30	0.333

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	37.9	26.6	36.9	26.3	36.0	26.2	28.1	34.3	27.7	33.7	27.4	33.2	1.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.6	22.3	30.4	26.2	21.8	30.0	26.0	21.5	29.8	90.3	34.6	88.5	34.0	87.3	33.4	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.8	4.1	3.4	DB	15.6	39.3	1.6	1.0	14.4	40.0	13.5	40.6	12.6	41.2	11.4	41.9	(4)
(5)			WB	12.9	29.4	1.6	1.1	11.7	30.1	10.7	30.8	9.8	31.4	8.6	32.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	28.9	26.9	28.8	30.4	31.1	30.5	29.8	29.0	28.9	28.6	28.5	27.7	26.5	(6)	
	DBStd	2.10	2.13	2.06	1.73	1.63	1.50	1.26	1.20	1.09	1.10	1.15	1.69	2.18	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)	
	CDD10.0	6895	524	528	633	632	635	593	591	586	559	574	531	510	(10)	
	CDD18.3	3853	266	294	375	382	377	343	332	328	309	316	281	252	(11)	
	CDH23.3	44279	2829	3499	4776	4947	4649	4054	3715	3605	3274	3421	2977	2533	(12)	
	CDH26.7	20126	1278	1756	2546	2694	2294	1813	1494	1413	1211	1333	1235	1058	(13)	
(14) Wind	WSAvg	0.8	0.7	0.8	0.8	0.8	0.7	0.8	0.8	0.8	0.6	0.6	0.9	1.0	(14)	
(15) Precipitation	PrecAvg	1329	7	25	53	92	144	166	176	199	243	181	38	5	(15)	
	PrecMax	1815	30	107	148	216	324	316	349	352	480	325	113	33	(16)	
	PrecMin	945	0	0	4	19	33	76	74	76	77	79	0	0	(17)	
	PrecStd	184	8	29	47	53	65	62	61	68	82	72	35	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.2	37.1	38.8	39.6	38.5	36.7	35.4	34.8	34.6	34.5	34.9	34.8	(19)	
		MCWB	24.7	25.2	25.8	26.9	27.3	27.0	27.0	27.2	27.1	26.6	26.0	25.1	(20)	
	2%	DB	34.3	36.1	37.6	38.2	36.9	35.3	34.2	33.9	33.6	33.6	34.0	33.7	(21)	
		MCWB	24.3	25.1	25.7	26.9	27.2	27.0	26.8	26.9	26.9	26.5	25.7	24.6	(22)	
	5%	DB	33.4	35.1	36.5	37.0	35.6	34.3	33.4	33.1	32.7	32.9	33.0	32.7	(23)	
		MCWB	23.9	24.9	25.7	26.8	27.1	26.9	26.6	26.7	26.7	26.3	25.4	24.1	(24)	
	10%	DB	32.4	34.1	35.4	35.7	34.4	33.3	32.4	32.2	31.8	32.1	32.1	31.7	(25)	
		MCWB	23.4	24.6	25.6	26.6	27.0	26.8	26.4	26.5	26.4	26.1	25.1	23.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.1	27.4	28.3	28.6	28.2	27.9	28.4	27.9	27.7	27.2	26.6	(27)	
		MCDB	33.0	34.7	35.2	36.4	35.6	33.7	33.3	33.0	32.8	32.4	32.7	32.4	(28)	
	2%	WB	25.7	26.4	26.9	27.7	28.0	27.7	27.3	27.6	27.3	27.2	26.7	25.7	(29)	
		MCDB	31.8	33.7	34.6	35.5	34.6	33.1	32.5	32.2	32.0	31.9	31.9	31.6	(30)	
	5%	WB	25.0	25.9	26.5	27.3	27.6	27.3	27.0	27.1	27.0	26.8	26.2	25.1	(31)	
		MCDB	30.9	32.9	33.9	34.8	33.8	32.6	32.1	31.7	31.6	31.4	31.2	30.6	(32)	
	10%	WB	24.4	25.4	26.1	27.0	27.2	27.0	26.6	26.7	26.6	26.5	25.8	24.5	(33)	
		MCDB	30.1	31.8	33.3	34.1	33.0	32.2	31.2	31.0	30.8	30.8	30.4	29.9	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	11.0	10.5	9.9	8.4	7.6	7.1	7.1	6.9	7.3	8.5	9.9	(35)	
		MCDBR	11.6	11.6	11.8	11.2	9.8	8.7	8.3	8.0	7.9	8.2	9.4	10.5	(36)	
		MCWBR	3.8	3.0	2.7	2.8	2.7	2.4	2.5	2.6	2.6	2.7	3.0	3.4	(37)	
	5% WB	MCDBR	10.2	10.6	10.7	10.4	8.9	8.0	7.7	7.6	7.4	7.6	8.4	9.5	(38)	
		MCWBR	3.7	3.0	2.9	3.0	2.8	2.6	2.7	2.8	2.7	2.8	2.9	3.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.546	0.626	0.642	0.575	0.488	0.470	0.465	0.468	0.488	0.530	0.484	0.476	(40)	
	taud		1.926	1.760	1.767	1.968	2.251	2.304	2.315	2.289	2.223	2.076	2.190	2.163	(41)	
	Ebn at Noon		754	712	714	762	817	823	830	836	825	781	806	807	(42)	
	Edh at Noon		187	229	232	189	140	131	130	135	145	164	142	144	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.97	5.15	5.24	5.58	5.33	5.05	4.82	4.78	4.59	4.70	4.85	4.82	(44)	
	RadStd		0.30	0.23	0.23	0.24	0.28	0.20	0.29	0.30	0.23	0.29	0.31	0.19	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.29	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)		N/A	N/A	N/A	N/A	+0.22	+0.20	N/A	N/A	N/A	N/A

Nomenclature: See separate page